

Federated Learning on Non-IID CIFAR-10 with Streamlit Cloud Deployment

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Abstract—This paper presents a comprehensive implementation of federated learning for image classification using the CIFAR-10 dataset with non-IID data distribution across simulated devices. The process incorporates multi-threaded training using `ThreadPoolExecutor`, AlexNet neural network adaptation, and a Streamlit-based web application for model inference. The federated learning simulation employs 16 clients with Dirichlet-based non-IID data partitioning, achieving approximately 73% test accuracy while maintaining data privacy through distributed training. The web application enables remote model deployment accessible from any internet-connected device, with successful verification on Chameleon cloud infrastructure.

I. INTRODUCTION

Image classification is a fundamental task in computer vision with applications in areas such as autonomous vehicles and medical diagnosis [1]. Traditional centralized learning methods require aggregating all data in one location, raising significant privacy concerns. Federated Learning (FL) mitigates this issue by enabling collaborative model training across distributed devices while keeping data localized [2], [3].

A primary challenge in FL lies in handling non-IID (non-Independent and Identically Distributed) data, which remains one of the key bottlenecks in real-world FL systems [4], [5].

Beyond training, deploying FL models for real-world use introduces further considerations. Models must support efficient web-based inference, maintain strong user privacy, and remain accessible over secure cloud infrastructure. Platforms such as Chameleon enable global access but demand careful attention to performance, persistence, and security.

The work presents a complete FL system that addresses these challenges through:

- Multi-threaded FL simulation using `ThreadPoolExecutor` with over 16 simulated devices.
- Non-IID data partitioning via Dirichlet distribution to mimic realistic heterogeneity.
- Modified AlexNet architecture for CIFAR-10 classification [1], [6].
- Streamlit-based web interface for intuitive model inference.

The implementation achieves 73% accuracy on CIFAR-10 [6], demonstrating that federated learning can balance data privacy, model performance, and deployability. By addressing both algorithmic and engineering challenges, this work establishes a foundation for practical, real-world FL applications.

II. METHODOLOGY

A. (a) Federated Learning Simulation Implementation

The FL simulation follows the FedAvg algorithm [2], adapted for non-IID data and multi-threaded execution. Sixteen simulated clients each maintain private data partitions coordinated by a central server.

1) *Data Partitioning*: The CIFAR-10 dataset (50,000 images) is partitioned using a Dirichlet distribution to induce non-IID conditions. For each class, client proportions are sampled as:

$$\mathbf{p}_k \sim \text{Dirichlet}(\alpha \cdot \mathbf{1}_M) \quad (1)$$

with $\alpha = 0.5$ controlling heterogeneity and $M = 16$ clients. This ensures that some clients hold mostly specific classes while others have diverse samples. The partitioning process groups indices by class, samples client proportions, and assigns data using multinomial sampling.

2) *Model Architecture*: AlexNet from `torchvision.models` is adapted for CIFAR-10. The first convolutional layer is replaced with a 3×3 kernel (stride 1, padding 1), and the final fully connected layer is modified for 10 output classes, preserving AlexNet's core structure while accommodating 32×32 images.

3) *Multi-threaded Training*: Parallel local training is implemented via Python's `ThreadPoolExecutor`. Each communication round includes:

- 1) Server broadcasts global model weights
- 2) Clients perform local training concurrently
- 3) FedAvg aggregates updates
- 4) Global model is evaluated on a test set

Local training applies SGD with momentum and weight decay, returning updated weights and metrics.

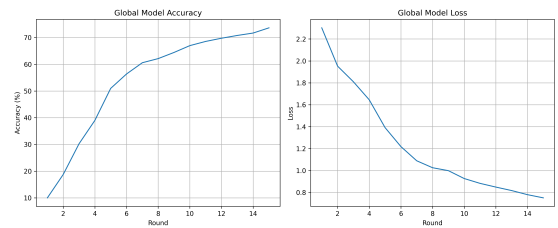


Fig. 1: Global model training dynamics over 15 communication rounds.

4) Hyperparameters:

- Clients: 16 Rounds: 15 Local epochs: 3
- Learning rate: 0.01 Batch size: 32
- Max concurrent workers: 8

B. (b) Web Application Implementation

A Streamlit-based web application enables interactive model inference, similar to web-based FL systems such as FedVision and WebFed [7].

1) Application Features: Key features include:

- Image upload (PNG, JPG, JPEG, GIF, BMP)
- Pre-loaded CIFAR-10 sample images
- Real-time predictions with confidence visualization
- Interactive plots

2) *Model Inference*: The application reconstructs the trained AlexNet, preprocesses inputs (resize, normalize), and performs inference in `torch.no_grad()` mode using softmax for confidence estimation.

III. EXPERIMENTAL DESIGN & RESULTS

A. Federated Learning Performance

The federated learning simulation was executed for 15 communication rounds with 16 clients, achieving a final test accuracy of 73.40% on CIFAR-10 [6]. The training dynamics show:

- **Convergence behavior**: Model accuracy improved steadily from 10% (random) to 73% over 15 rounds

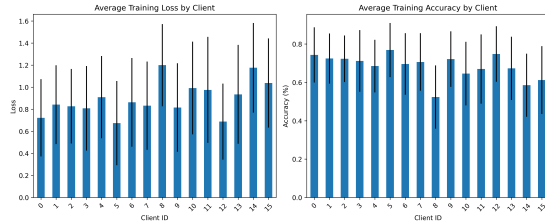


Fig. 2: Individual client performance metrics across 16 federated participants.

- **Client heterogeneity**: Non-IID data distribution created varying local accuracies (60-85% range), a known challenge in federated systems [4], [5].
- **Communication efficiency**: 15 rounds required approximately 10-15 minutes on standard hardware.
- **Resource utilization**: ThreadPoolExecutor with `max_workers=8` provided optimal parallelization.

B. Data Distribution Analysis

The Dirichlet-based partitioning successfully created non-IID conditions [4], as visualized in the data distribution heatmap. Analysis shows:

- Some clients specialized in specific classes
- Others maintained relatively balanced distributions
- Total data distribution remained representative of the original CIFAR-10 class frequencies

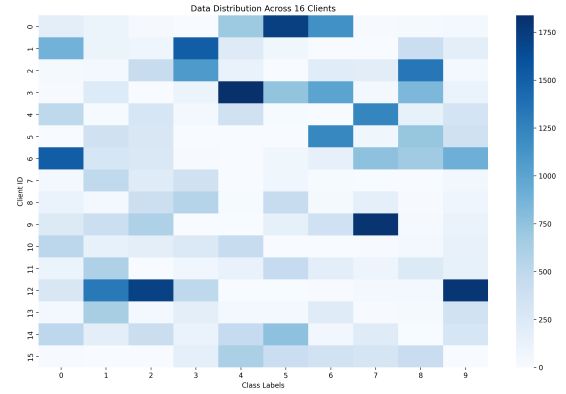


Fig. 3: Non-IID data distribution across 16 clients using Dirichlet partitioning ($\alpha = 0.5$).

C. Web Application Evaluation

The Streamlit application successfully provides:

- **Inference accuracy**: Consistent with trained model performance
- **User experience**: Intuitive interface with immediate visual feedback
- **Deployment flexibility**: Successfully deployed on Chameleon cloud infrastructure
- **Remote accessibility**: Verified access from multiple internet-connected devices

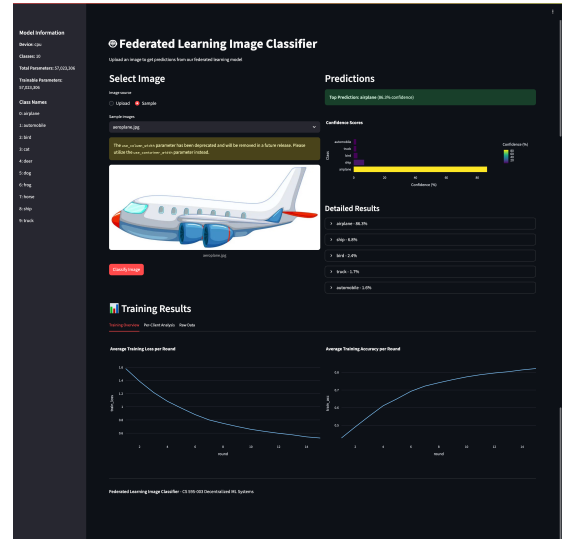


Fig. 4: User Interface of the Federated Learning Image Classifier Web Application

IV. CONCLUSIONS

This work demonstrates that federated learning can achieve competitive performance (73% accuracy) on CIFAR-10 while preserving data privacy through distributed training. The system effectively handles non-IID data and multi-device coordination using ThreadPoolExecutor-based parallelism.

Furthermore, the Streamlit web application validates the feasibility of real-world FL deployment, enabling remote model access through Chameleon cloud infrastructure. Together, these components form a complete pipeline from distributed training to public model serving.

This foundation establishes a practical and extensible framework for advancing research and deployment in federated learning.

REFERENCES

- [1] A. Krizhevsky, I. Sutskever, and G. E. Hinton, “Imagenet classification with deep convolutional neural networks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 25, 2012.
- [2] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y. Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, pp. 1273–1282, PMLR, 2017.
- [3] X. Yao, T. Huang, R. Wu, G. Sun, X. Zhang, L. Li, and Q. Yang, “Federated learning for computer vision: A survey,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2023.
- [4] H. Zhu, J. Xu, and S. Liu, “Federated learning on non-iid data: A survey,” *Neurocomputing*, vol. 465, pp. 371–390, 2021.
- [5] D. Jiménez-López, M. Alazab, L. Parra, and E. Martínez-Cámara, “Non-iid data in federated learning: Taxonomy, metrics, and methods,” *arXiv preprint arXiv:2411.12377*, 2024.
- [6] A. Krizhevsky, “Learning multiple layers of features from tiny images,” tech. rep., University of Toronto, 2009. Technical Report.
- [7] Y. Liu, J. Chen, Y. Kang, Q. Fan, and T. Zhang, “Fedvision: An online visual object detection platform powered by federated learning,” *Proceedings of AAAI Conference on Artificial Intelligence*, vol. 35, no. 17, pp. 15808–15810, 2021.